Introduction to TCGA portal

Nelly Selem & Victor Muñiz

https://portal.gdc.cancer.gov/

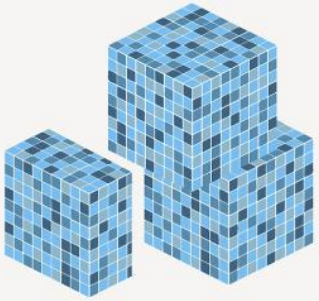
THE CANCER GENOME ATLAS (TCGA) BY THE NUMBERS

TCGA produced over

2.5

PETABYTES

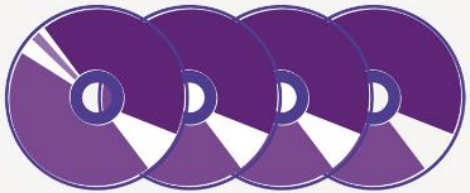
of data



To put this into perspective, **1 petabyte** of data is equal to

212,000

DVDs



TCGA data describes



33

DIFFERENT TUMOR TYPES

...including

10

RARE CANCERS

...based on paired tumor and normal tissue sets collected from



11,000

PATIENTS

...using

7

DIFFERENT DATA TYPES



RESULTS & FINDINGS



MOLECULAR BASIS OF CANCER

Improved our understanding of the genomic underpinnings of cancer

For example, a TCGA study found the basal-like subtype of breast cancer to be similar to the serous subtype of ovarian cancer on a molecular level, suggesting that despite arising from different tissues in the body, these subtypes may share a common path of development and respond to similar therapeutic strategies.



TUMOR SUBTYPES

Revolutionized how cancer is classified

TCGA revolutionized how cancer is classified by identifying tumor subtypes with distinct sets of genomic alterations.*



THERAPEUTIC TARGETS

Identified genomic characteristics of tumors that can be targeted with currently available therapies or used to help with drug development

TCGA's identification of targetable genomic alterations in lung squamous cell carcinoma led to NCI's Lung-MAP Trial, which will treat patients based on the specific genomic changes in their tumor.

THE TEAM



20

COLLABORATING INSTITUTIONS

across the United States and Canada

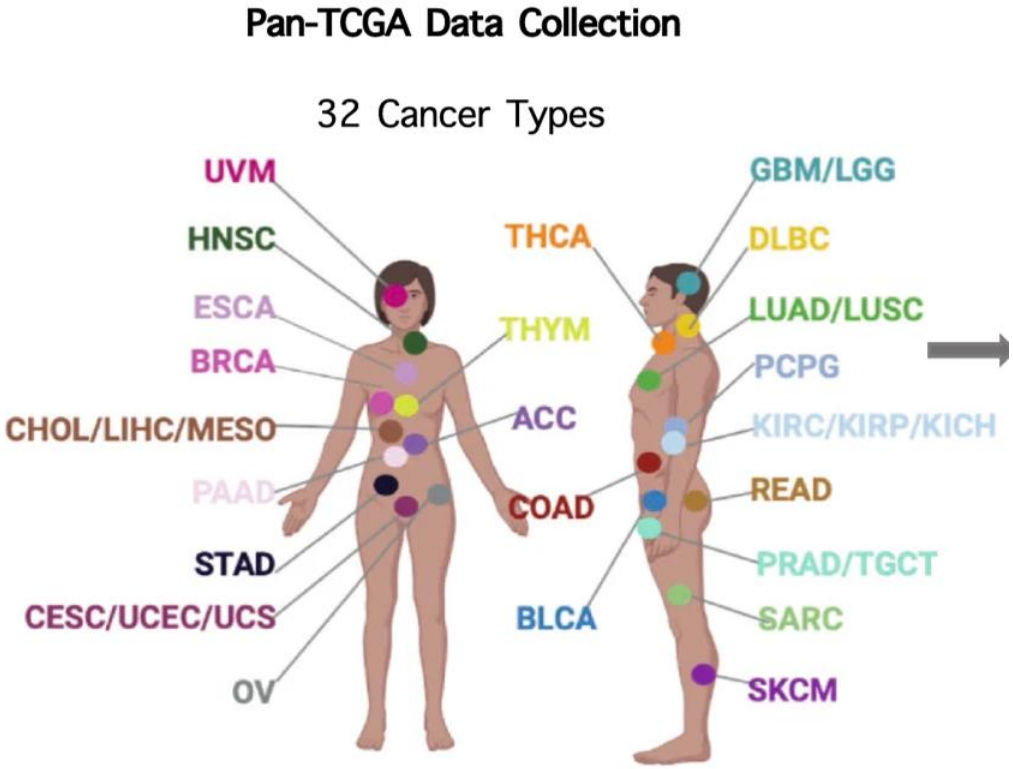
WHAT'S NEXT?

The Genomic Data Commons (GDC) houses TCGA and other NCI-generated data sets for scientists to access from anywhere. The GDC also has many expanded capabilities that will allow researchers to answer more clinically relevant questions with increased ease.

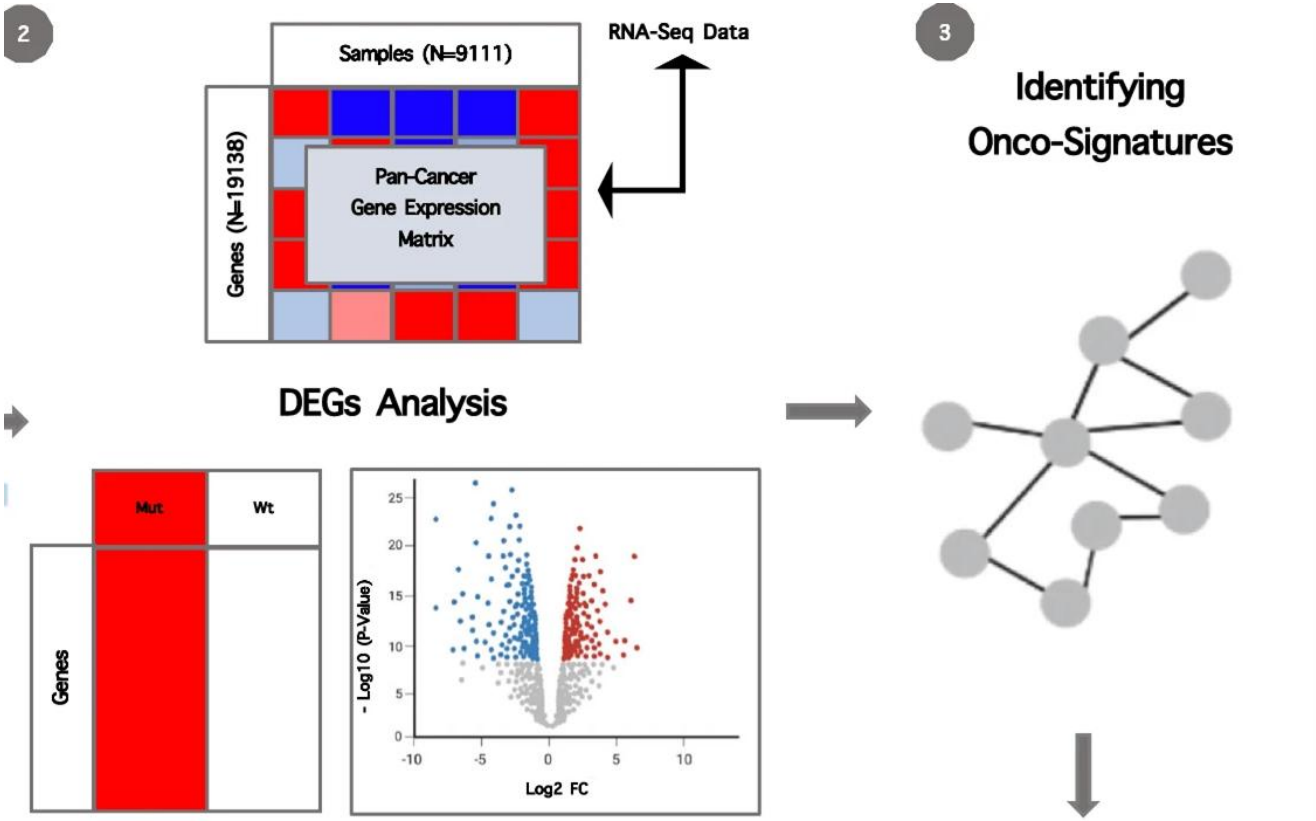


*TCGA's analysis of stomach cancer revealed that it is not a single disease, but a disease composed of four subtypes, including a new subtype characterized by infection with Epstein-Barr virus.

The pan cancer data collection contains 32 cancer types



Diferential Expression Analysis helps to define 105 onco-signatures



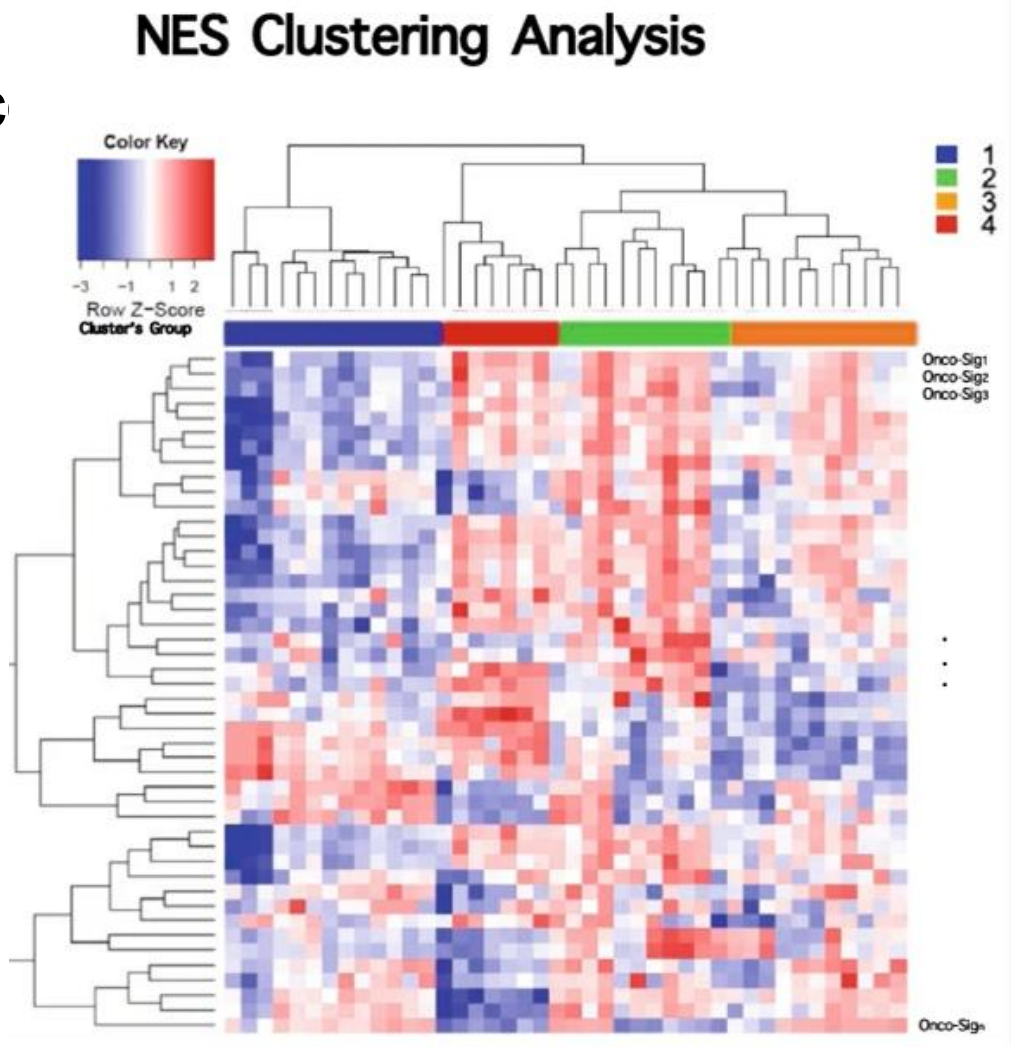
Normalized Enrichment Sc is calculated from onco - signature

U= Mann-Whitney
n number of genes
m number of reference genes

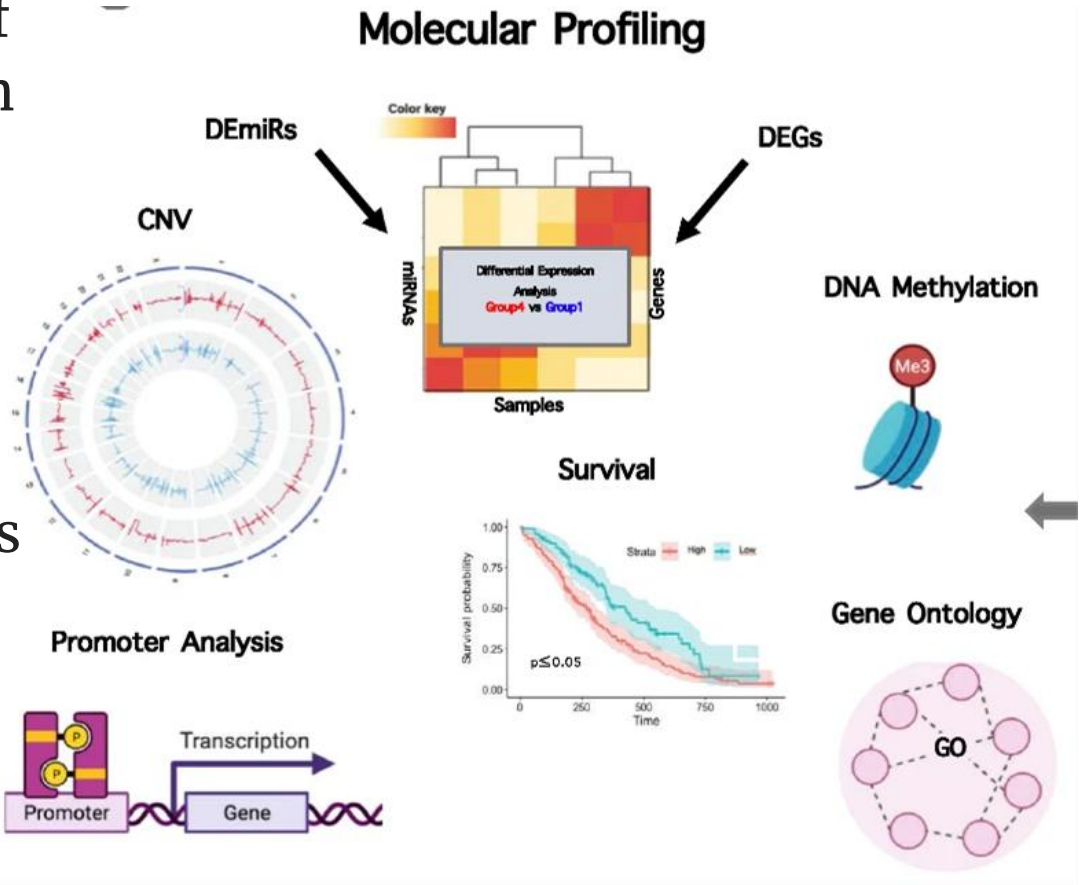
ssMWW-GST
in TCGA-BRCA

$$NES = 1 - \frac{U}{mn}$$

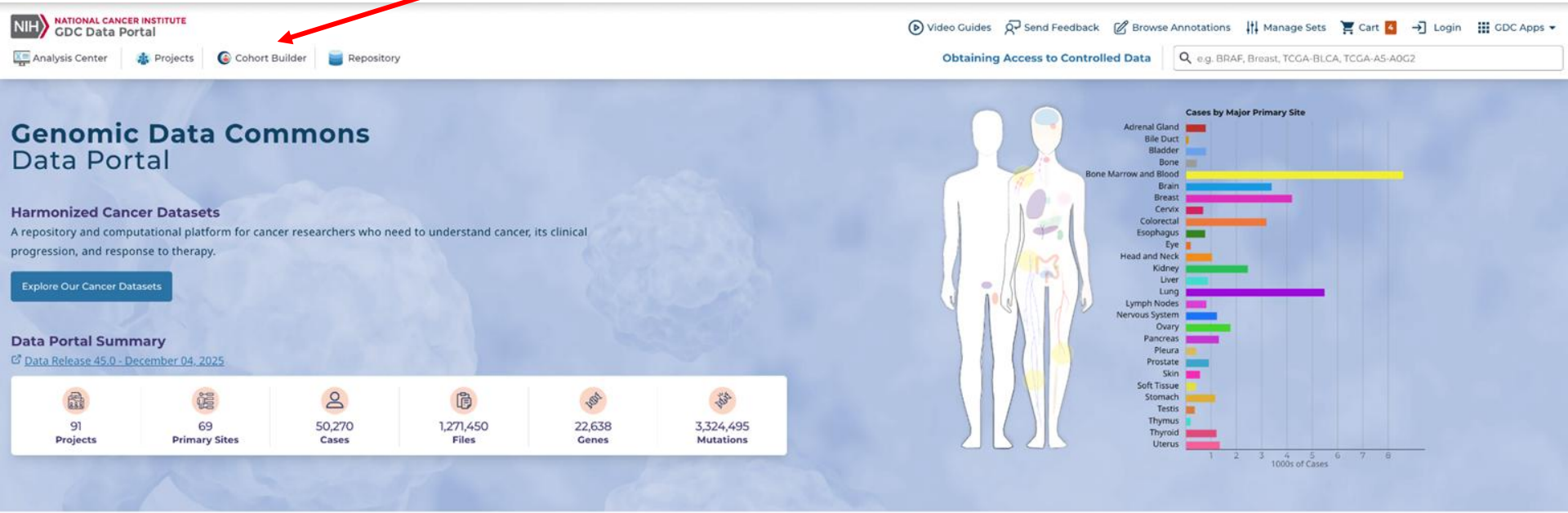
5 Prognostic Onco-Signatures Selection



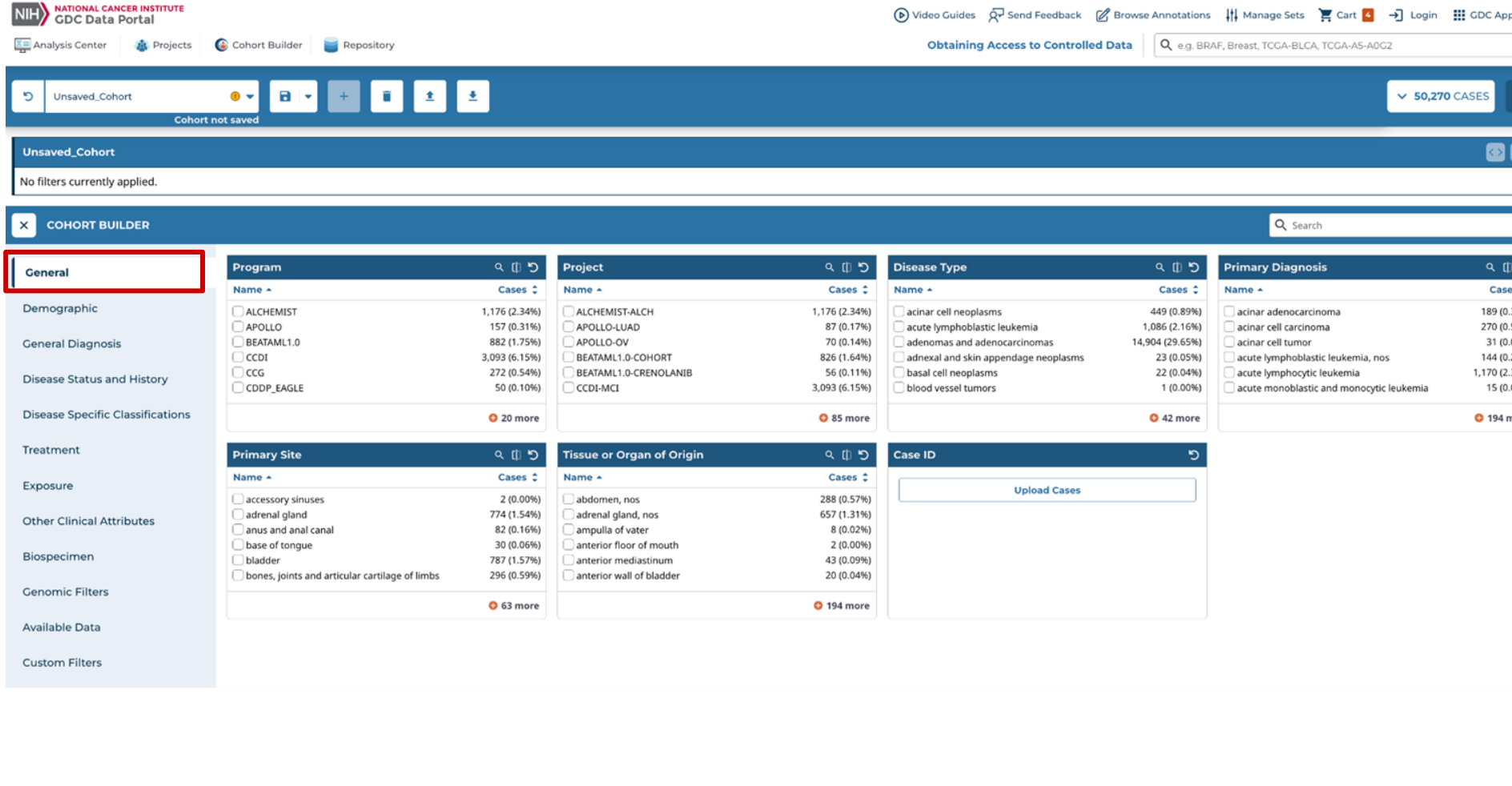
Molecular characterization of the identified groups through survival analysis, differential expression analysis, copy number variation analysis, DNA methylation analysis, promoter analysis, gene ontology enrichment analysis

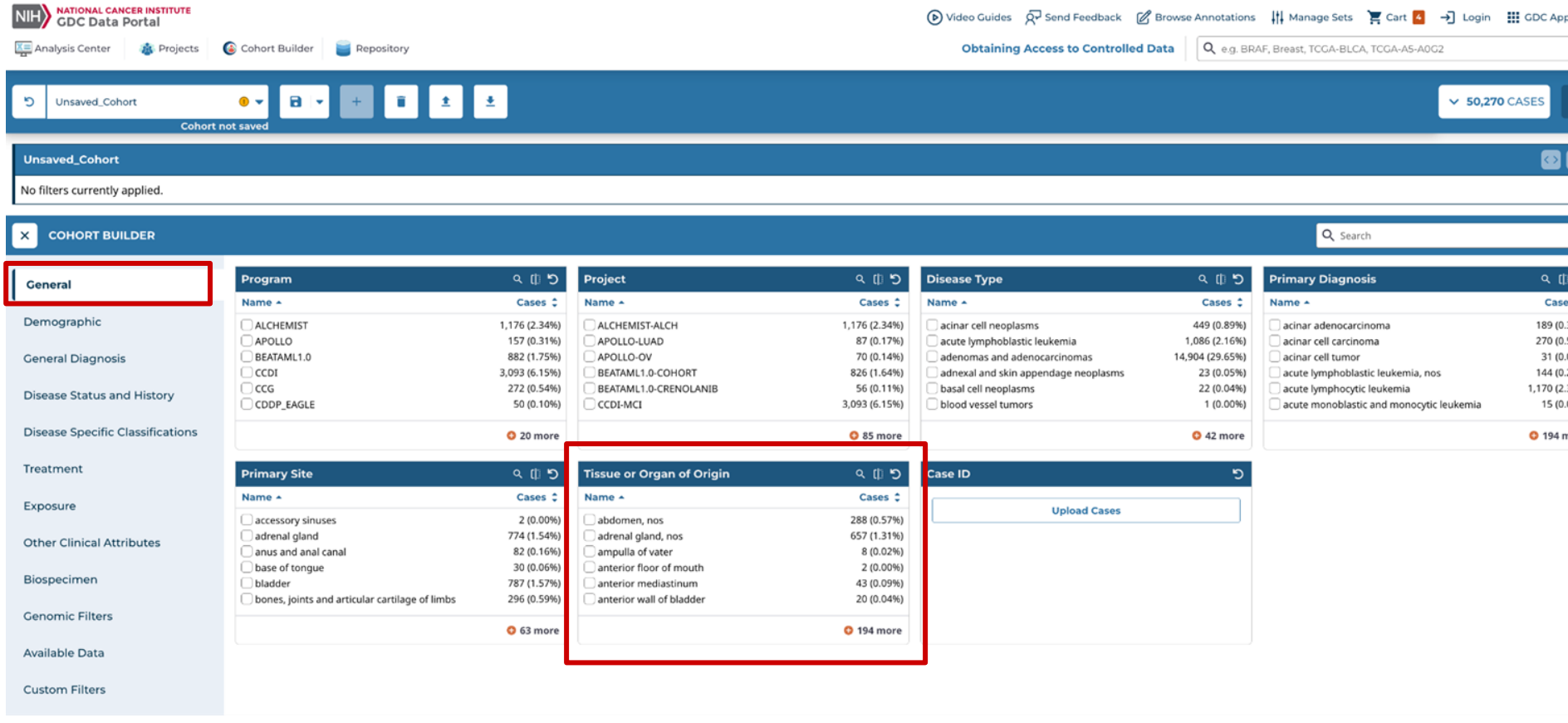


Lets go to the portal !

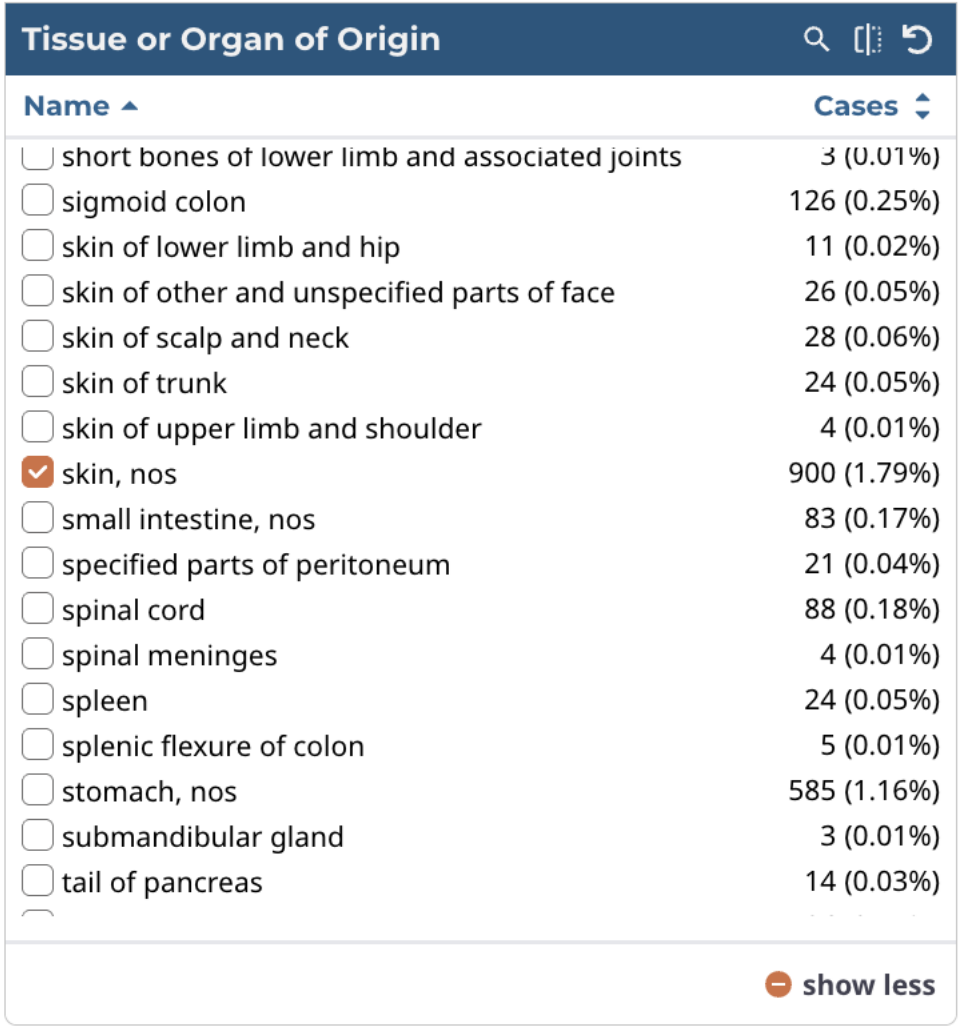


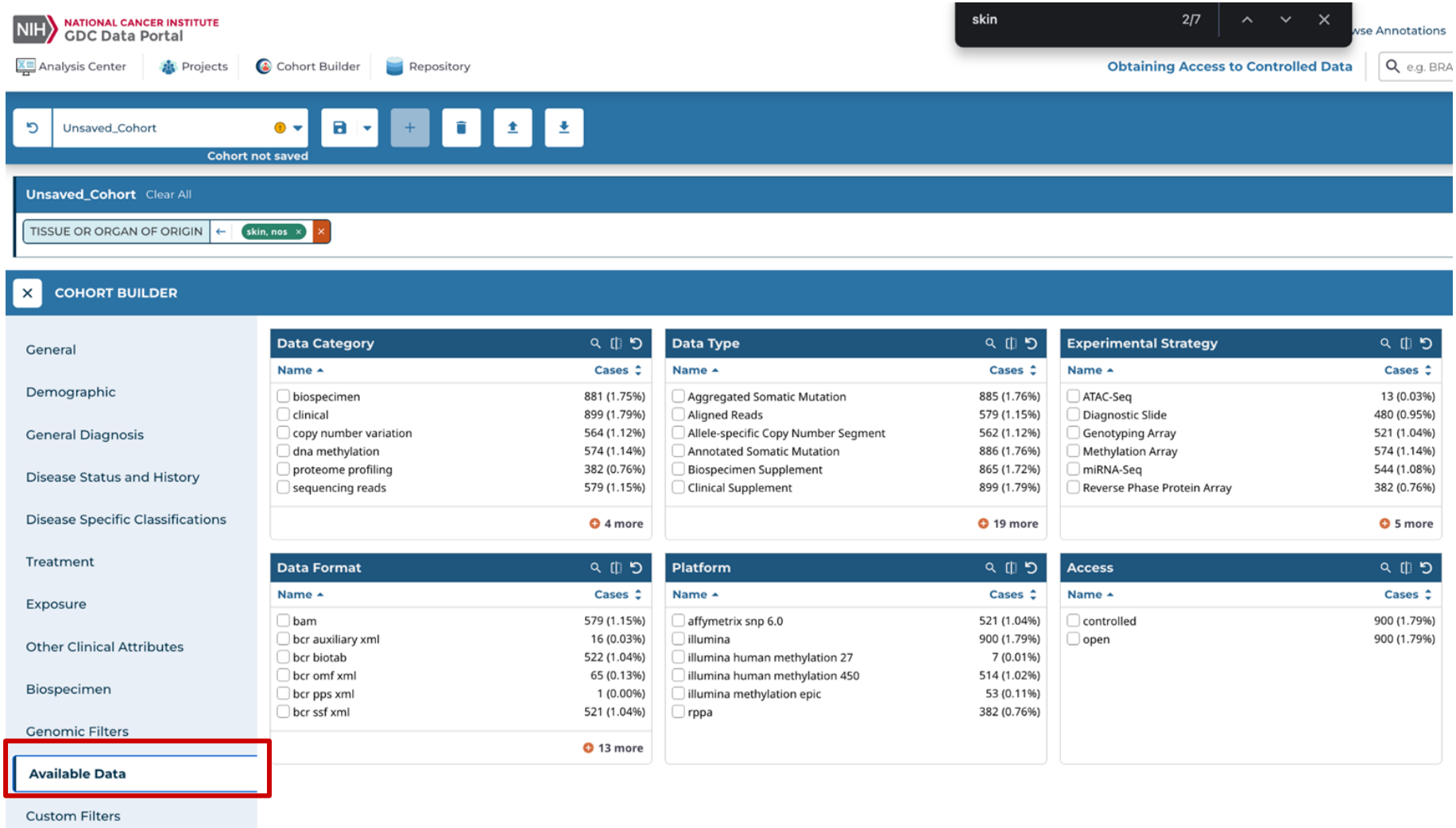
High-quality Datasets
From Foundational Cancer Genomic Studies


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Login
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Analysis Center
Projects
Cohort Builder
Repository
Obtaining Access to Controlled Data
e.g. BRAF, Breast, TCGA-BLCA, TCGA-A5-A0G2
Unsaved_Cohort
Cohort not saved
50,270 CASES
Unsaved_Cohort
No filters currently applied.
COHORT BUILDER
Search
General
Demographic
General Diagnosis
Disease Status and History
Disease Specific Classifications
Treatment
Exposure
Other Clinical Attributes
Biospecimen
Genomic Filters
Available Data
Custom Filters
Program
Name
Cases
ALCHEMIST 1,176 (2.34%)
APOLLO 157 (0.31%)
BEATAML1.0 882 (1.75%)
CCDI 3,093 (6.15%)
CCG 272 (0.54%)
CDDP_EAGLE 50 (0.10%)
20 more
Project
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APOLLO-LUAD 87 (0.17%)
APOLLO-OV 70 (0.14%)
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BEATAML1.0-CRENOLANIB 56 (0.11%)
CCDI-MCI 3,093 (6.15%)
85 more
Disease Type
Name
Cases
acinar cell neoplasms 449 (0.89%)
acute lymphoblastic leukemia 1,086 (2.16%)
adenomas and adenocarcinomas 14,904 (29.65%)
adnexal and skin appendage neoplasms 23 (0.05%)
basal cell neoplasms 22 (0.04%)
blood vessel tumors 1 (0.00%)
42 more
Primary Diagnosis
Name
acinar adenocarcinoma
acinar cell carcinoma
acinar cell tumor
acute lymphoblastic leukemia, nos
acute lymphocytic leukemia
acute monoblastic and monocytic leukemia
Primary Site
Name
Cases
accessory sinuses 2 (0.00%)
adrenal gland 774 (1.54%)
anus and anal canal 82 (0.16%)
base of tongue 30 (0.06%)
bladder 787 (1.57%)
bones, joints and articular cartilage of limbs 296 (0.59%)
63 more
Tissue or Organ of Origin
Name
Cases
abdomen, nos 288 (0.57%)
adrenal gland, nos 657 (1.31%)
ampulla of vater 8 (0.02%)
anterior floor of mouth 2 (0.00%)
anterior mediastinum 43 (0.09%)
anterior wall of bladder 20 (0.04%)
194 more
Case ID
Upload Cases


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Upload Cases

Select skin non specific




skin 2/7
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e.g. BRA
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Unsaved_Cohort Clear All
TISSUE OR ORGAN OF ORIGIN
skin, nos
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Demographic
General Diagnosis
Disease Status and History
Disease Specific Classifications
Treatment
Exposure
Other Clinical Attributes
Biospecimen
Genomic Filters
Available Data
Custom Filters
Data Category
Name
Cases
biospecimen 881 (1.75%)
clinical 899 (1.79%)
copy number variation 564 (1.12%)
dna methylation 574 (1.14%)
proteome profiling 382 (0.76%)
sequencing reads 579 (1.15%)
4 more
Data Type
Aggregated Somatic Mutation 885 (1.76%)
Aligned Reads 579 (1.15%)
Allele-specific Copy Number Segment 562 (1.12%)
Annotated Somatic Mutation 886 (1.76%)
Biospecimen Supplement 865 (1.72%)
Clinical Supplement 899 (1.79%)
19 more
Experimental Strategy
ATAC-Seq 13 (0.03%)
Diagnostic Slide 480 (0.95%)
Genotyping Array 521 (1.04%)
Methylation Array 574 (1.14%)
miRNA-Seq 544 (1.08%)
Reverse Phase Protein Array 382 (0.76%)
5 more
Data Format
bam 579 (1.15%)
bcr auxiliary xml 16 (0.03%)
bcr biotab 522 (1.04%)
bcr omf xml 65 (0.13%)
bcr pps xml 1 (0.00%)
bcr ssf xml 521 (1.04%)
13 more
Platform
affymetrix snp 6.0 521 (1.04%)
illumina 900 (1.79%)
illumina human methylation 27 7 (0.01%)
illumina human methylation 450 514 (1.02%)
illumina methylation epic 53 (0.11%)
rppa 382 (0.76%)
Access
controlled 900 (1.79%)
open 900 (1.79%)

Unsaved_Cohort

Cohort not saved

Unsaved_Cohort Clear All

TISSUE OR ORGAN OF ORIGIN ← skin, nos × | EXPERIMENTAL STRATEGY ← RNA-Seq ×

COHORT BUILDER

- General
- Demographic
- General Diagnosis
- Disease Status and History
- Disease Specific Classifications
- Treatment
- Exposure
- Other Clinical Attributes
- Biospecimen
- Genomic Filters
- **Available Data**
- Custom Filters

Data Category

Name	Cases
sequencing reads	572 (1.14%)
structural variation	571 (1.14%)
transcriptome profiling	572 (1.14%)

Data Type

Name	Cases
Aligned Reads	572 (1.14%)
Gene Expression Quantification	572 (1.14%)
Splice Junction Quantification	572 (1.14%)
Transcript Fusion	571 (1.14%)

Experimental Strategy

Name	Cases
ATAC-Seq	13 (0.03%)
Diagnostic Slide	480 (0.95%)
Genotyping Array	521 (1.04%)
Methylation Array	574 (1.14%)
miRNA-Seq	544 (1.08%)
Reverse Phase Protein Array	382 (0.76%)
RNA-Seq	572 (1.14%)
Targeted Sequencing	323 (0.64%)
Tissue Slide	546 (1.09%)
WGS	541 (1.08%)
WXS	577 (1.15%)

show less

Data Format

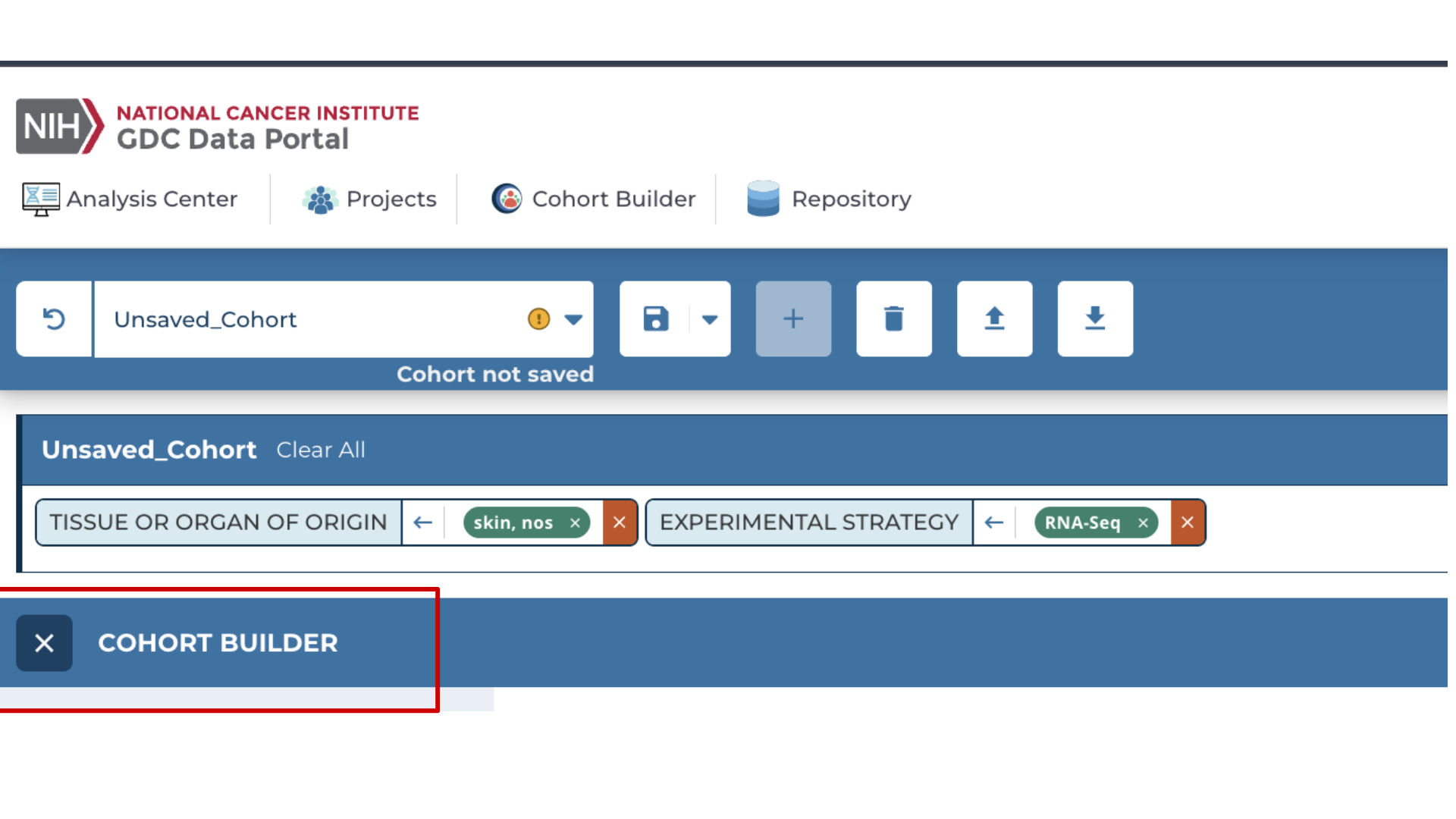
Name	Cases
bam	572 (1.14%)
bedpe	571 (1.14%)
tsv	572 (1.14%)

Platform

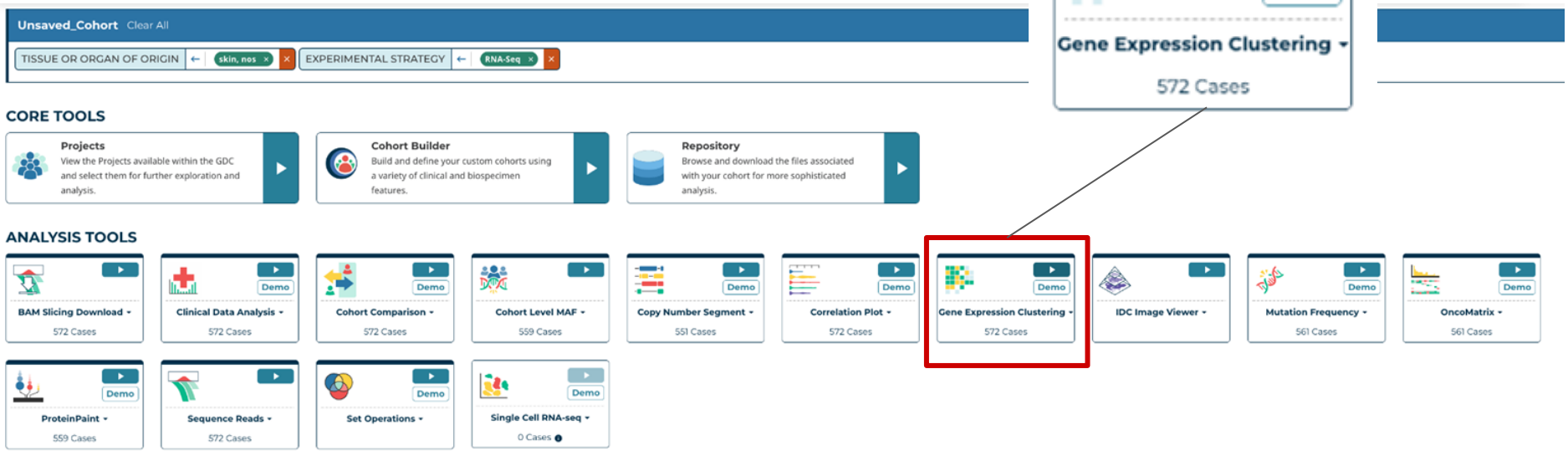
Name	Cases
illumina	572 (1.14%)

Access

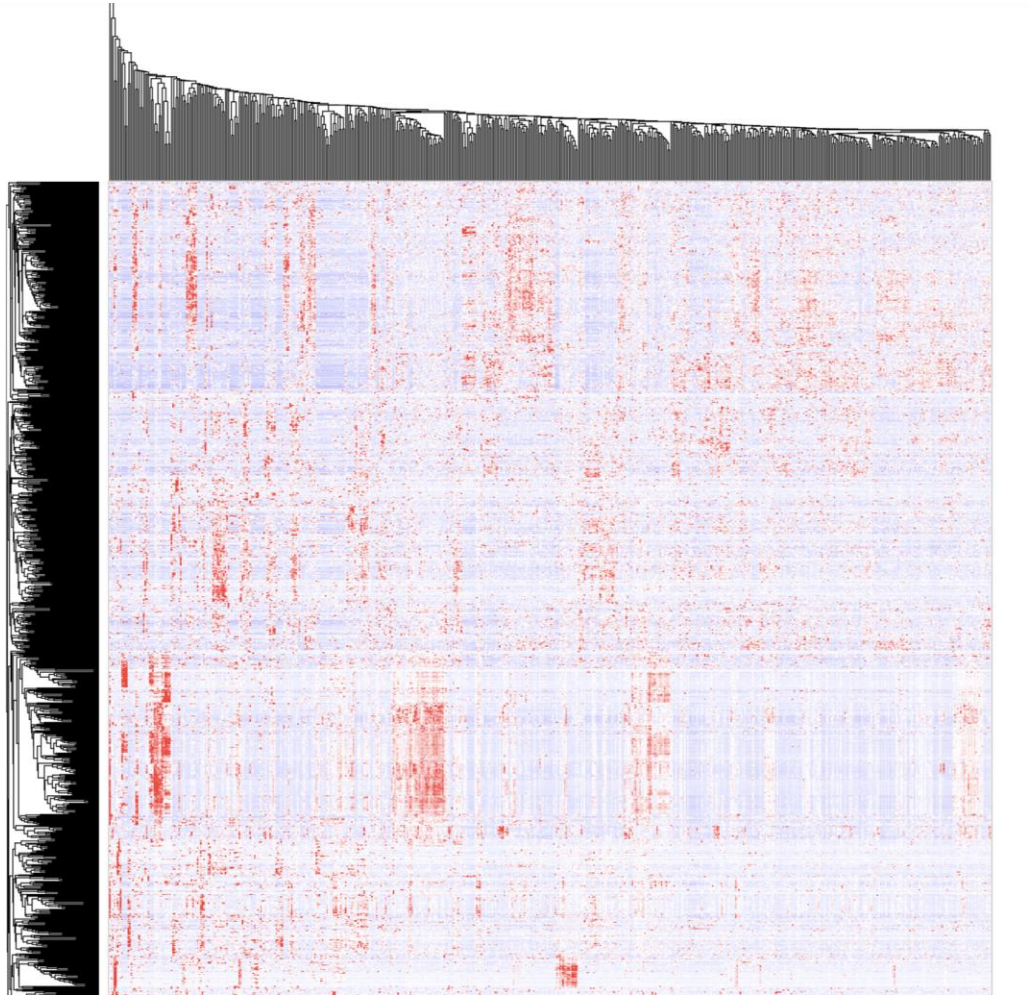
Name	Cases
controlled	572 (1.14%)
open	572 (1.14%)


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TISSUE OR ORGAN OF ORIGIN
skin, nos
EXPERIMENTAL STRATEGY
RNA-Seq
COHORT BUILDER

Select gene expresion, or explore



Here is the diferencial expression matrix



Download the tsv data !